

Vandermeer Securities

The screen said \$327,000 of free money on a \$50 million trade, and Joaquín Ferreira had learned to treat that sentence as a description of something he did not yet understand.

Ferreira ran a small relative-value book at Vandermeer Securities. On a Tuesday morning his monitor flagged a three-year corporate note — an on-the-run issue from a single-A industrial, 6 percent annual coupon, three years to maturity — quoted at 103.20. Vandermeer's own curve, bootstrapped from the par yields of the same issuer's benchmark bonds, said the note should be worth 103.85.

The gap was 65 cents per hundred of face. On the \$50 million block that a dealer had offered, that was \$327,000.

The desk had two ways to think about it. The first was that the note was cheap and should be bought. The second was that the curve was wrong, or that something about the note made it different from the bonds the curve was built from, in which case there was no gap at all and the trade was a way of paying \$327,000 to learn something.

Ferreira's head of desk, Ingeborg Aalto, had a standing rule about relative-value trades: before you put on the position, write down the trade that extracts the value with no market risk at all, and then say why that trade is not already done by someone with a lower cost of capital than yours.

The stripped-security market for this issuer's paper existed but was thin. The note itself was on the run and had traded 40 times in the past week. Two of the benchmark bonds used in the bootstrap had not traded in nine days, and their quotes were dealer indications. The note was also, Ferreira noticed on the second pass, trading special in the repo market at 55 basis points below general collateral.

EXHIBIT 1 — PAR YIELDS, SAME ISSUER

MATURITY	PAR YIELD
1 year	4.00%
2 years	4.35%
3 years	4.60%
4 years	4.78%
5 years	4.90%

Annual coupons, annual compounding. Par yields are the coupon rates at which the issuer's bonds of each maturity would price at 100.

EXHIBIT 2 — THE NOTE

Coupon	6.00%, annual
Maturity	3 years
Quoted price	103.20
Block offered	\$50m face
Repo	trading 55bp special

ASSIGNMENT

- 1 Bootstrap the spot rates for years 1 through 5 from the par yields. Report the discount factors and the spot rates.
- 2 Price the three-year 6 percent note off the bootstrapped curve. Confirm the 103.85 figure and the size of the gap on \$50 million.
- 3 Compute the note's yield to maturity at the quoted price of 103.20. Compare it to the three-year spot rate and to the three-year par yield. Explain why all three differ, and which comparison is meaningful.
- 4 Compute the implied one-year forward rates for years 2 through 5. Is the market forecasting rising short rates? Answer carefully.
- 5 Write the answer to Aalto's rule. Describe the risk-free trade that extracts the 65 cents, and then give three reasons the gap might be real compensation rather than an error. Recommend.